

COMP 532

Machine Learning and Bioinspired Optimisation

Lecture 5 Reinforcement Learning



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Recap

- Reinforcement Learning
 - Roots of RL
 - Key features of RL
 - Elements of RL
- Games
 - Tic-Tac-Toe example

Overview

- Problem Solving from Nature
- Reinforcement Learning
 - Roots of RL
 - Key features of RL
 - Elements of RL
 - Multi-armed bandits
- Deep Learning
- Multi-Agent Learning
- Swarm Intelligence

Bandit



Multi-armed bandit problem

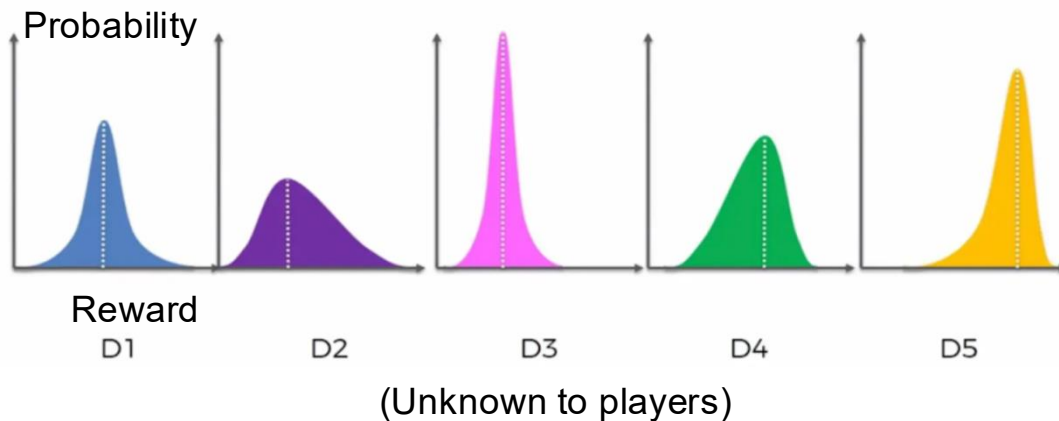


Multi-armed bandit problem



- Scenario: Pull machine $k \Rightarrow$ sample from unknown reward distribution $D_k \Rightarrow$ observe reward
- Problem: Given a finite number of pulls T , how can I optimize my winnings?
- How much should I explore?
How much should I exploit?

Multi-armed bandit problem



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Bandits are the simplest RL setting

Reinforcement learning studies **sequential decision making under uncertainty**:

- An agent interacts with an environment
- Actions influence rewards
- The agent must learn from evaluative feedback

Bandits are the simplest RL setting:

- No state
- No transition dynamics
- Only action $\rightarrow$ reward

The Multi-Armed Bandit Problem

Setting:

- A finite set of actions (arms):

$$\mathcal{A} = \{1, 2, \dots, K\}$$

- Each action a is associated with an unknown reward distribution D_a

Interaction protocol (time step t):

- Agent selects action $A_t \in \mathcal{A}$
- Receives reward: $R_t \sim D_{A_t}$

Objective:

- Maximise expected cumulative reward over T steps: $\max \mathbb{E}[\sum_{t=1}^T R_t]$

Action-Value Estimates

True Action Values

- The **true value** of an action is its expected reward: $Q^*(a) = \mathbb{E}[R_t \mid A_t = a]$

Key properties:

- $Q^*(a)$ is fixed but unknown
- The agent must estimate it from experience
- At time step t , the agent maintains estimates: $Q_t(a) \approx Q^*(a)$
- These estimates are used to guide action selection.
- Learning in bandits = **learning accurate action-value estimates.**

Exploration vs Exploitation

- **Exploitation:** choose the action with highest estimated value
- **Exploration:** choose other actions to gain information

Greedy action at time t : $a_t^* = \arg \max_a Q_t(a)$

Core dilemma:

- Exploitation improves short-term reward
- Exploration improves long-term knowledge

Stationary vs Non-Stationary Bandits

Stationary:

- $Q^*(a)$ does not change over time

Non-stationary:

- $Q^*(a)$ may change

Examples of non-stationarity:

- User preferences
- Advertising fatigue
- Changing environments
- Implication:
- Exploration must *continue* in non-stationary settings

Action-Value Methods

- Action-value methods update estimates based only on observed rewards.
- If action a has been selected k times with rewards $r_1, \dots, r_k$: $Q_t(a) = \frac{1}{k} \sum_{i=1}^k r_i$
- By the Law of Large Numbers: $Q_t(a) \rightarrow Q^*(a)$ as $k \rightarrow \infty$

Incremental Update Rule

To avoid storing all past rewards:

- $Q_{t+1}(a) = Q_t(a) + \alpha_t(R_t - Q_t(a))$

Interpretation:

- New Estimate = Old Estimate + Step Size $\times$ (Target – Old Estimate)

This update form appears throughout reinforcement learning:

- Temporal-difference learning
- Q-learning
- Policy evaluation

Incremental Update Rule

Update Rule

- $Q_{t+1}(a) = Q_t(a) + \alpha_t(R_t - Q_t(a))$

Step Sizes

- **Sample-average step size** (stationary case): $\alpha_t = \frac{1}{N_t(a)}$
- **Constant step size** (non-stationary case): $0 < \alpha \leq 1$

Constant step sizes yield:

- Exponentially recency-weighted averages
- Faster adaptation to change

ϵ -Greedy Action Selection

With probability $1 - \epsilon$:

- Choose greedy action

With probability ϵ :

- Choose an action uniformly at random
- Formally:

$$A_t = \begin{cases} \arg \max_a Q_t(a) & \text{with prob. } 1 - \epsilon \\ \text{random action} & \text{with prob. } \epsilon \end{cases}$$

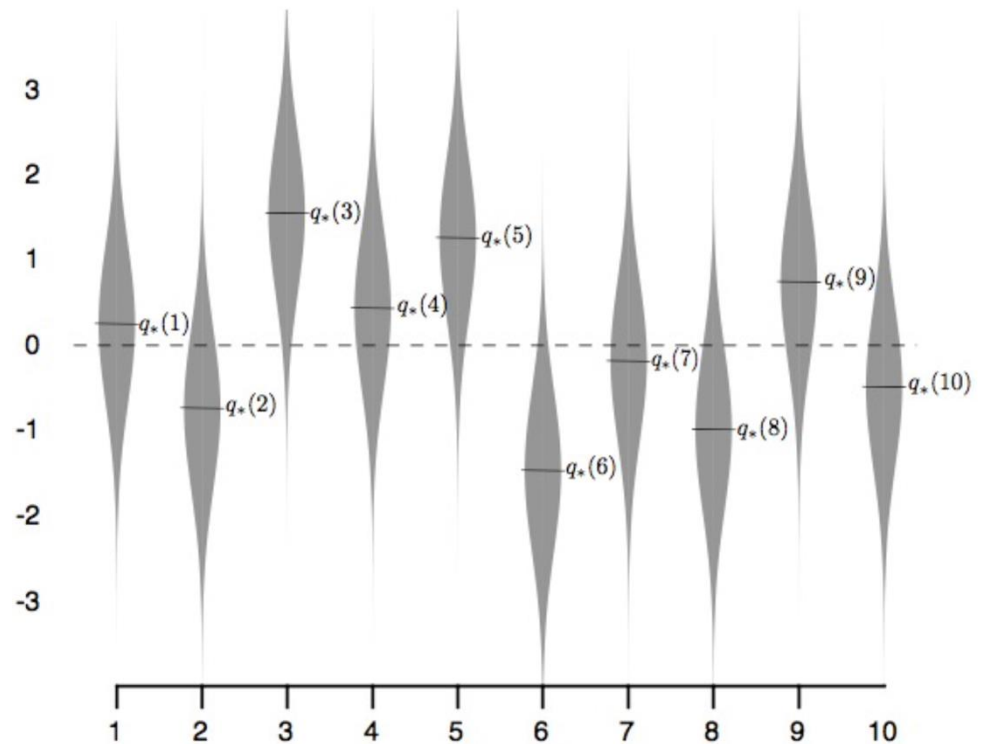
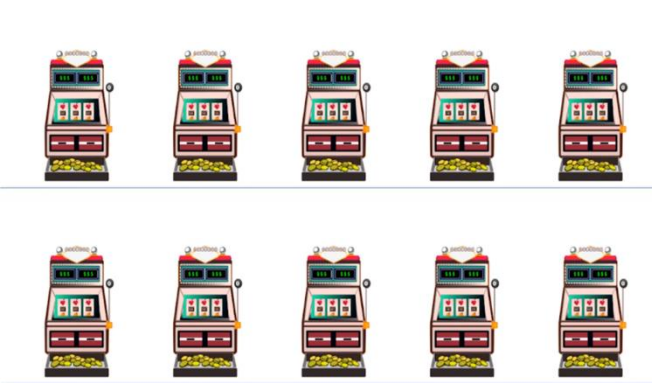
 Exploration

Softmax (Boltzmann) Action Selection

- Actions are selected probabilistically: $P(A_t = a) = \frac{\exp(Q_t(a)/\tau)}{\sum_b \exp(Q_t(b)/\tau)}$
- Temperature τ :
- $\tau \rightarrow 0$: greedy behaviour
- $\tau \rightarrow \infty$: uniform random
- Softmax trades off exploration smoothly.

Example: 10-Armed Testbed

- $n = 10$ possible arms/actions
- For each arm, there is a $Q^*(a)$. (It is chosen randomly and unknown to players.)



Note here we use $q_*(a)$ to indicate $Q^*(a)$.

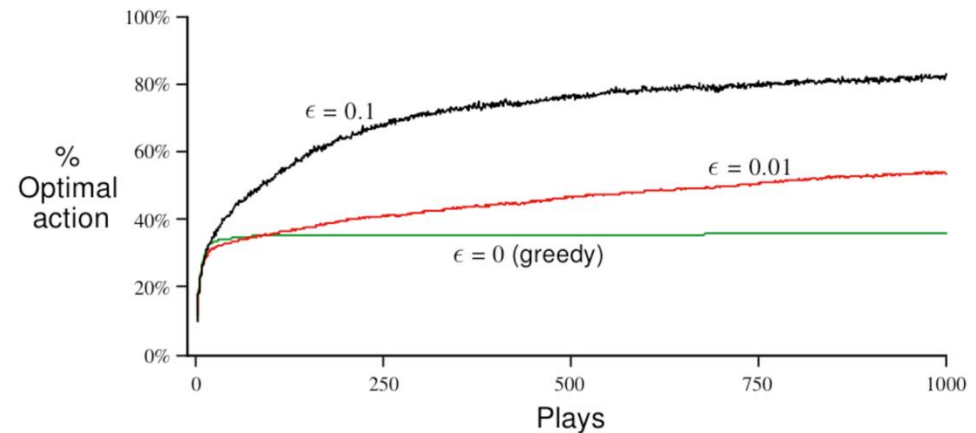
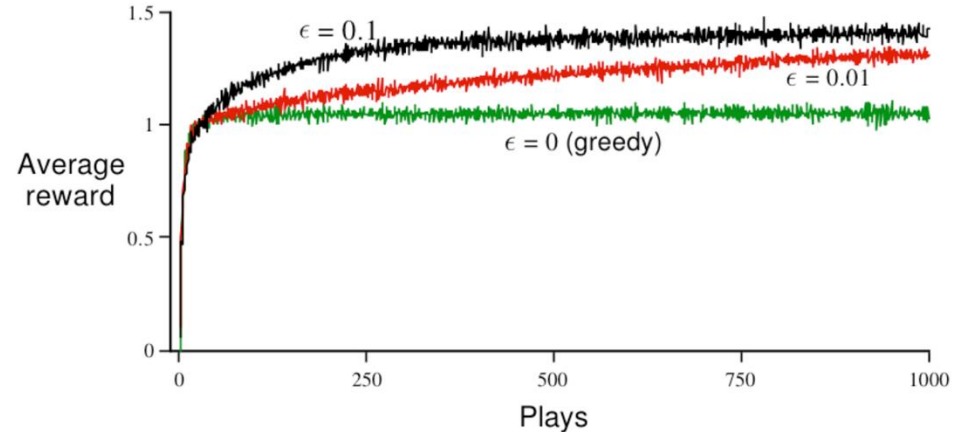
Different Methods on the 10-Armed Testbed

Evaluation protocol:

- 1000 time steps per run
- 1000 independent runs
- Average reward and % optimal action reported

Key result:

- Pure greedy performs poorly
- Moderate exploration yields better long-term performance



Summary

In this lecture, we learned:

- The formal definition of the multi-armed bandit problem
- Why exploration is unavoidable
- How to estimate action values incrementally
- ϵ -greedy and softmax exploration strategies
- The role of stationarity assumptions

Bandits form the conceptual foundation for full reinforcement learning with states and dynamics.